JOHANSSON ENGINEERING

Equipment Life Journal — PV-001 Compressed Air Receiver

Document No.	JDS-LOG-MEC-004
Revision	A
Date	2026-03-25
Status	EXAMPLE
Author	Nils Johansson

1. Equipment Identity

FIELD	VALUE
Vessel ID	PV-001
Description	Compressed air receiver, vertical
Manufacturer	Atlas Copco
Model	LT 500
Serial No.	AC-2018-4472
Year of manufacture	2018
Date commissioned	2018-09-14
Location	Göteborg Workshop — Compressor room
Design standard	SS-EN 13445
PED Category	III
AFS 2017:3 Class	A
Design pressure (PS)	11 bar
Design temperature	–10°C to +80°C
Volume	500 L
PS × V	5500 bar·L
Medium	Compressed air (Group 2)
Material	P265GH (shell), S235JR (saddles)
Wall thickness (nominal)	8.0 mm
Corrosion allowance	1.0 mm
Min. retirement thickness	6.2 mm
Expected service life	25 years (to 2043)
Relief valve	SV-001 — set at 11.5 bar

2. Lifecycle Timeline

DATE	AGE	EVENT	DESCRIPTION	BY
2018-09-14	0 yr	Commissioning	New vessel installed, hydrostatic test 16.5 bar witnessed by Kiwa Inspecta	Atlas Copco / KI
2018-09-14	0 yr	First inspection	Initial external and internal inspection passed	Kiwa Inspecta
2019-03-10	0.5 yr	Maintenance	Condensate drain valve replaced (original valve leaking)	N. Johansson
2020-09-15	2 yr	Scheduled inspection	External inspection — no findings	Kiwa Inspecta
2021-06-01	2.7 yr	Modification	Automatic condensate drain timer installed (15 min cycle)	N. Johansson
2022-09-18	4 yr	Scheduled inspection	External inspection — minor surface rust on bottom saddle, wire-brushed and painted	Kiwa Inspecta
2023-06-15	4.8 yr	Scheduled inspection	Internal revision — no corrosion, wall thickness 7.9 mm (UT)	Kiwa Inspecta
2023-06-15	4.8 yr	Relief valve test	SV-001 tested, lifts at 11.4 bar — within tolerance	Kiwa Inspecta
2024-03-22	5.5 yr	Maintenance	Pressure gauge replaced (failed calibration check)	N. Johansson
2025-06-15	6.8 yr	Scheduled inspection	Internal revision — wall thickness 7.8 mm (UT), 0.1 mm loss over 2 years	Kiwa Inspecta
2025-06-15	6.8 yr	Relief valve test	SV-001 tested, lifts at 11.5 bar — OK	Kiwa Inspecta

3. Wall Thickness Trend

DATE	AGE (YR)	LOCATION	MEASURED (MM)	LOSS FROM NOMINAL	ANNUAL RATE
2018-09-14	0	Bottom shell	8.0	—	—
2023-06-15	4.8	Bottom shell	7.9	0.1 mm	0.021 mm/yr
2025-06-15	6.8	Bottom shell	7.8	0.2 mm	0.029 mm/yr

Projected retirement: At current corrosion rate (0.029 mm/yr), minimum thickness of 6.2 mm will be reached in approximately **55 years** from manufacture (2073). Well within the 25-year expected service life.

Assessment: Corrosion rate is low and stable. No accelerated degradation detected. Continue monitoring at 2-year intervals.

4. Maintenance Record

DATE	WORK ORDER	DESCRIPTION	PARTS	DURATION	BY
2019-03-10	WO-2019-008	Replace condensate drain valve	1× ball valve DN15 PN16	1.5 hr	N. Johansson
2021-06-01	WO-2021-014	Install automatic drain timer	Timer unit, solenoid valve, tubing	3 hr	N. Johansson
2024-03-22	WO-2024-005	Replace pressure gauge	1× Wika 233.50 0–16 bar	0.5 hr	N. Johansson

5. Certificates & Documents

DOCUMENT	REFERENCE	DATE	HELD BY
EC Declaration of Conformity	Atlas Copco DoC-2018-4472	2018-08-20	Equipment file
Hydrostatic test certificate	KI-2018-HT-4472	2018-09-14	Equipment file
Material certificate 3.1 (shell)	Mill cert P265GH-2018-0442	2018-07	Equipment file
Inspection report 2023	KI-2023-IR-44721	2023-06-15	Equipment file
Inspection report 2025	KI-2025-IR-44721	2025-06-15	Equipment file
Relief valve test cert 2025	KI-2025-SV-001	2025-06-15	Equipment file

6. Next Actions

ACTION	DUE	RESPONSIBLE
External inspection	2027-06-15	Kiwa Inspecta
Internal revision + UT measurement	2027-06-15	Kiwa Inspecta
Relief valve SV-001 test	2027-06-15	Kiwa Inspecta
Condensate drain timer service	2026-06-01	N. Johansson

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-03-25	Nils Johansson	Example equipment life journal for JDS demonstration